

0.27

Avg Foul Delta All

-0.67

Avg Foul Delta Ghost

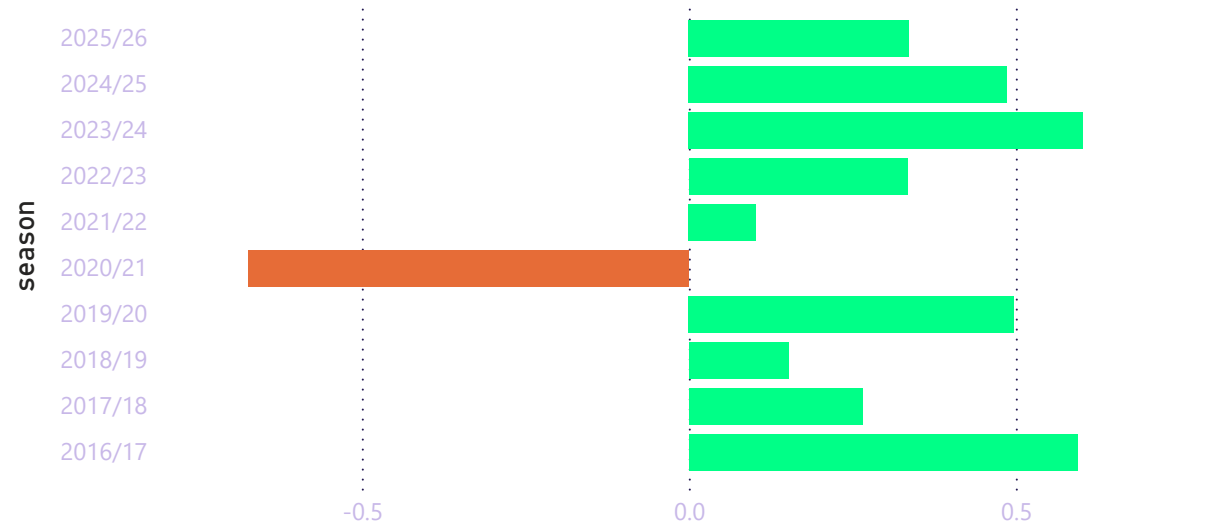
44.63

Home Win % All

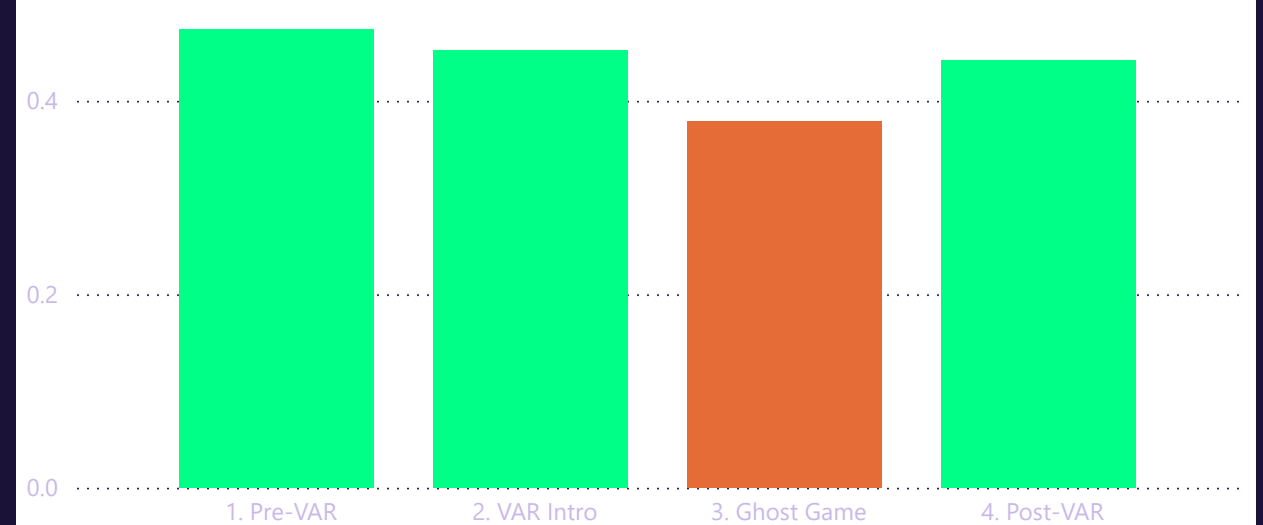
37.89

Home Win % Ghost

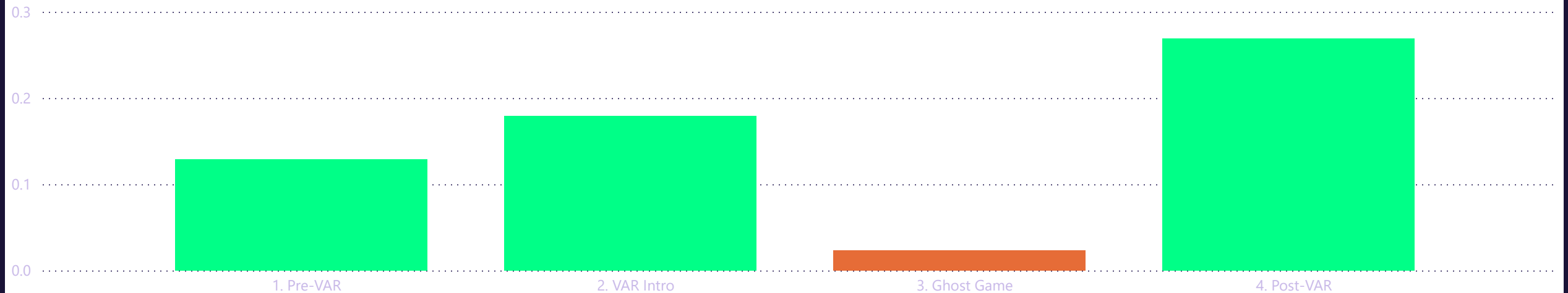
Foul Delta by Season (Away Fouls - Home Fouls)



Win rate at home by Era



Yellow Card Delta by Era (Away - Home)



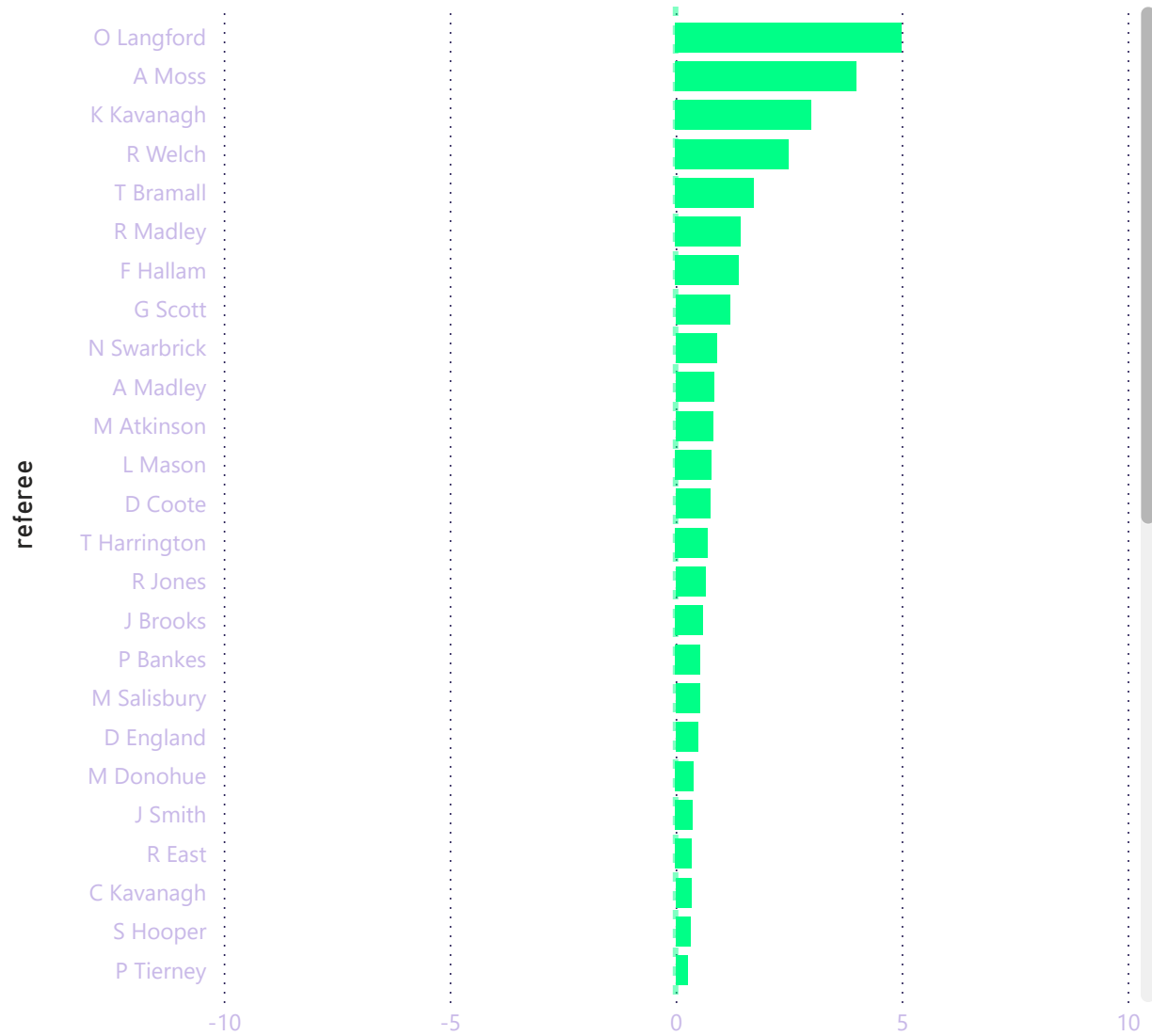
referee

All



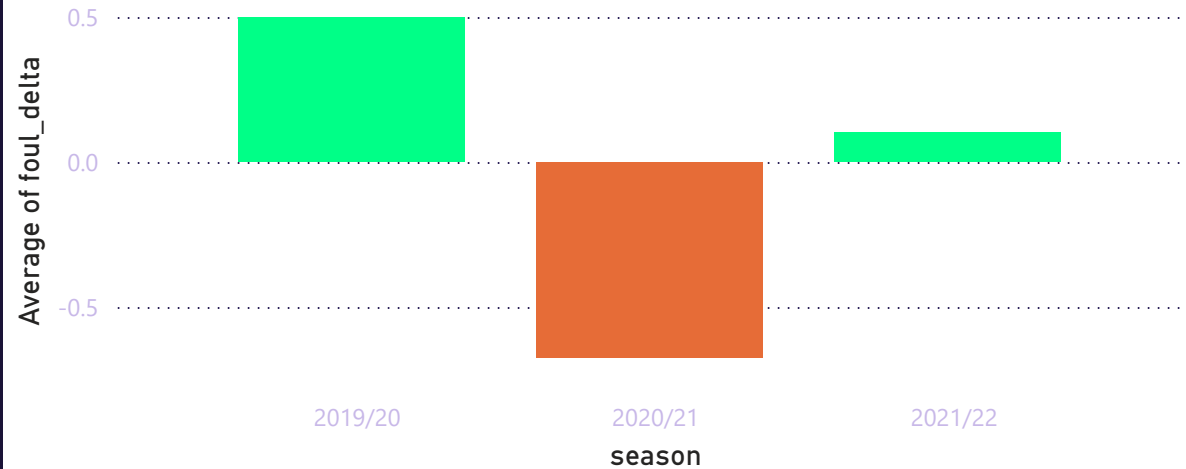
| referee     | Avg Foul Delta | Avg Yellow Delta | Home Win % | Matches Refereed |
|-------------|----------------|------------------|------------|------------------|
| A Taylor    | -0.26          | 0.04             | 43.05      | 295              |
| M Oliver    | 0.06           | 0.14             | 42.51      | 287              |
| C Pawson    | 0.10           | 0.18             | 41.92      | 229              |
| S Attwell   | -0.34          | 0.16             | 45.13      | 195              |
| C Kavanagh  | 0.34           | 0.22             | 44.56      | 193              |
| P Tierney   | 0.27           | 0.18             | 41.05      | 190              |
| M Atkinson  | 0.83           | 0.15             | 42.26      | 168              |
| A Marriner  | 0.15           | 0.09             | 37.20      | 164              |
| M Dean      | 0.14           | 0.02             | 43.83      | 162              |
| J Moss      | -0.16          | 0.03             | 43.75      | 160              |
| S Hooper    | 0.32           | 0.42             | 45.32      | 139              |
| K Friend    | 0.05           | -0.03            | 47.06      | 136              |
| A Madley    | 0.84           | 0.41             | 42.86      | 133              |
| P Bankes    | 0.55           | 0.62             | 53.91      | 115              |
| D Coote     | 0.77           | 0.18             | 46.43      | 112              |
| R Jones     | 0.66           | 0.63             | 47.27      | 110              |
| G Scott     | 1.20           | 0.26             | 55.00      | 100              |
| D England   | 0.49           | 0.60             | 42.42      | 99               |
| J Gillett   | -0.27          | 0.00             | 34.12      | 85               |
| L Mason     | 0.80           | 0.11             | 52.38      | 84               |
| J Brooks    | 0.61           | -0.22            | 52.70      | 74               |
| S Barrott   | -0.26          | 0.48             | 43.55      | 62               |
| M Salisbury | 0.54           | 0.08             | 49.18      | 61               |
| T Bramall   | 1.73           | 0.52             | 48.08      | 52               |
| R Madley    | 1.44           | 0.00             | 50.00      | 50               |
| T Robinson  | 0.16           | 0.57             | 48.98      | 49               |

Foul Delta by Referee by referee

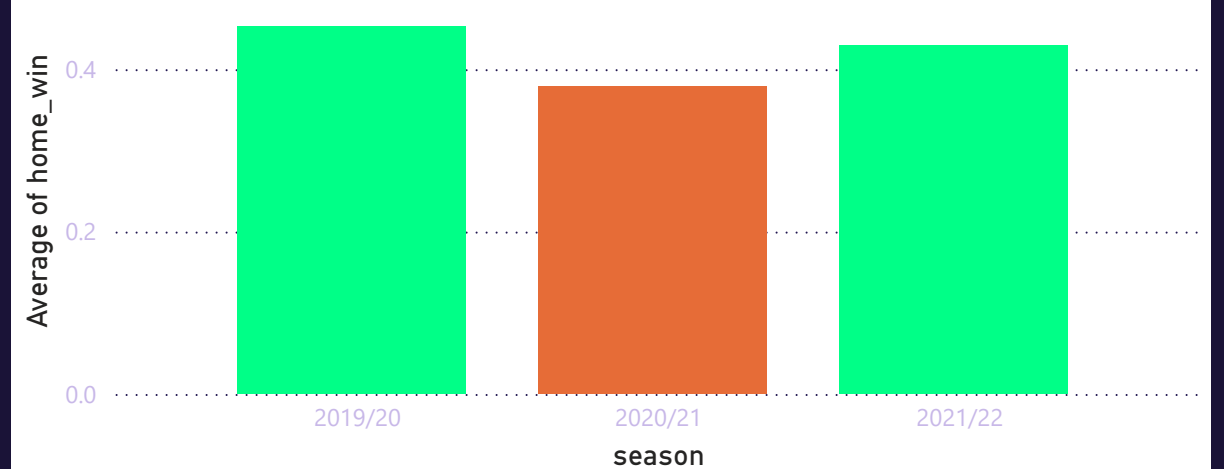


Foul Delta by Referee

Average of foul\_delta by season



Average of home\_win by season



## The Ghost Game Effect

In 2020/21, all EPL matches were played behind closed doors due to COVID-19. With no crowd pressure, referee decisions shifted measurably. Home foul delta dropped from +0.50 to -0.67, and home win rate fell from 45% to 38%. When fans returned in 2021/22, both metrics partially recovered. This natural experiment suggests that crowd noise influences referee decision-making in favour of the home team.